

Om Kumar Jha

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EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU) <i>Bachelor of Technology – Information Technology — GPA: 7.95</i>	Delhi, India <i>Aug 2024 – May 2028</i>
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SKILLS SUMMARY

- **Languages:** Python, C, C++, SQL
- **RAG & LLMs:** LangChain, ChromaDB, Sentence-Transformers, HuggingFace, Groq API, OpenRouter, Ollama
- **ML / AI:** Scikit-learn, TensorFlow, PyTorch, spaCy, FinBERT, YOLOv8
- **Cloud & Tools:** AWS EC2, Git, GitHub, Streamlit, Docker (Basics)

EXPERIENCE

Indian Navy (WESEE) <i>Data Science (Intern)</i>	Delhi, India <i>July 2025 - Sep 2025</i>
◦ Built a Python + SQL ingestion pipeline that fragmented high-frequency data streams and applied interval-based sampling, reducing downstream load by 70–90% and stabilizing real-time analytics ◦ Trained supervised learning models on high-resolution satellite data and validated them using train–test splits and cross-validation, improving detection consistency by 60–70% ◦ Developed ML workflows for geospatial datasets, including preprocessing, feature engineering, hyperparameter tuning, and model evaluation using accuracy/F1-score, improving model reliability and cutting analysis time by ~40% ◦ Designed and deployed a risk-scoring algorithm using 5+ operational parameters, improving threat-zone prediction accuracy and reducing route-planning time by ~30%	

PROJECTS

- **QueryForge** | *Python, LangChain, ChromaDB, Sentence-Transformers, Ollama, Streamlit:*
 - Built a multi-document RAG pipeline enabling semantic Q&A over large corpora using `all-MiniLM-L6-v2` embeddings, ChromaDB vector store, and Llama2 via Ollama as a fully local LLM backend
 - Implemented chunking strategies using LangChain RecursiveCharacterTextSplitter; designed retrieval evaluation using Precision@K, MRR, and F1-score to benchmark and iteratively improve retrieval quality
 - Built Streamlit UI for document upload, natural language querying, and highlighted answer extraction with source attribution
- **IssueRouter** | *Python, FastAPI, BART, spaCy, Sentence-Transformers, Groq API, React:*
 - Used `all-MiniLM-L6-v2` (Sentence-Transformers) to embed and semantically cluster citizen complaints from X (Twitter) into unified issue groups, reducing redundant reports reaching government officers
 - Implemented LLM summarization pipeline via Groq API (`llama3-8b-8192`) to generate structured officer-ready action briefs from raw unstructured tweet clusters
 - Built end-to-end ingestion-to-output pipeline: tweet stream → zero-shot classification → embedding → semantic clustering → LLM summarization → ranked officer dashboard
- **Digital Persona** | *Python, LangChain, FAISS, Sentence-Transformers, Groq API, Streamlit:*
 - Built a live-deployed RAG chatbot over LinkedIn profile data: Apify scraper → LangChain chunking → `all-MiniLM-L6-v2` embeddings → FAISS vector index → Groq LLM (`llama-3.3-70b-versatile`) for grounded multi-turn Q&A
 - Chose RAG over full-context injection to prevent token overflow on large profiles and ground answers strictly in retrieved profile data, reducing hallucination
 - Deployed on Streamlit Cloud with full error handling for private profiles, API rate limits, and LLM failures; zero LinkedIn credentials required via Apify actor

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification	Retrieval Augmented Generation (RAG)	AWS Educate
Machine Learning Foundations	Introducing Generative AI with AWS	Postman API Fundamentals
Scholar		McKinsey Forward

HONORS AND AWARDS

- **Harvard Project for Asian and International Relations (HPAIR)** 2025
 - Selected as a Delegate for HPAIR Asia Conference at The University of Tokyo through competitive global application
- **Trident Netra, Indian Navy (WESEE)** 2026
 - Contributed to Trident Netra, a naval AI system currently deployed and operational across Indian Navy vessels in real time
 - Project showcased at India AI Impact Summit 2026, Bharat Mandapam, New Delhi at the official Indian Navy stall